LDR

DETECT | RASPBERRY PI | LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses



Street light

DETECT How dark it is → **DECIDE** Is it darker than dusk? → **DO** Switch the lamp on



Biscuit tin alarm

DETECT Light inside the tin → **DECIDE** Is the lid open? → **DO** Sound the alarm



Window Light lands here

Track Zigzag sensor

Legs Either way round

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import LightSensor

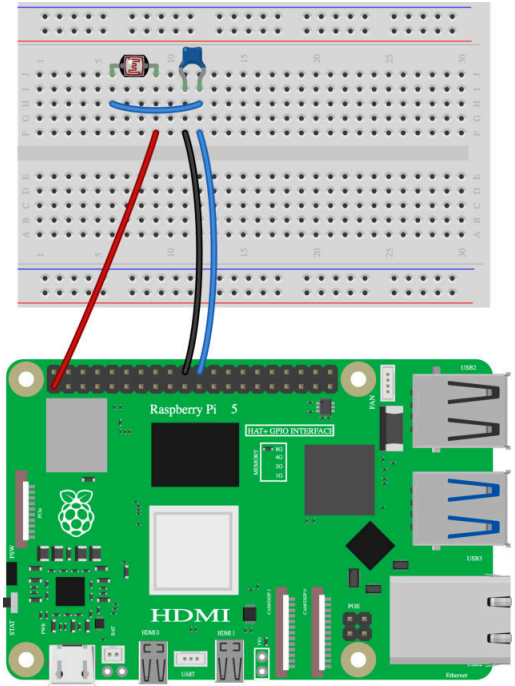
ldr = LightSensor(25)

light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light. Also `ldr.light_detected` and `ldr.when_light`.

Wiring

RASPBERRY PI 5



3V3 (pin 1) → LDR → **GPIO 25** (pin 22) → 1 µF → **GND** (pin 20)